

Focus Mate AI Dataset Documentation

Overview

This document describes the dataset used to train the Focus Mate attention detection AI model.

Dataset Information

Dataset Name

Synthetic Demo Dataset (for development and testing)

Location

- **Path:** data/processed/synthetic_demo/
- **Metadata:** data/processed/synthetic_demo/metadata.jsonl
- **Images:** data/processed/synthetic_demo/frames/

Dataset Size

- Default: 1,000 samples
 - Configurable via `--limit` parameter during generation
-

Dataset Generation

Generation Script

File: ml/data_ingest.py

Command to Generate

```
python ml/data_ingest.py ingest synthetic_demo --limit 1000
```

Generation Process

The synthetic dataset is generated programmatically using the `generate_synthetic_samples()` function in `ml/data_ingest.py` (lines 70-140).

For each sample:

1. Randomly selects an attention label
2. Generates realistic gaze angles based on the label
3. Creates corresponding head pose values
4. Assigns eye openness probability
5. Generates a synthetic image (160 × 120 RGB)
6. Saves metadata in JSONL format

Attention Labels (11 Classes)

The dataset includes 11 distinct attention states:

Label	Description	Typical Gaze Yaw	Typical Gaze Pitch	Eyes Open
focused	Looking directly at screen	$0^\circ \pm 2^\circ$	$0^\circ \pm 2^\circ$	0.8-1.0
looking_left	Gaze directed left	$-15^\circ \pm 3^\circ$	$0^\circ \pm 2^\circ$	0.7-1.0
looking_right	Gaze directed right	$+15^\circ \pm 3^\circ$	$0^\circ \pm 2^\circ$	0.7-1.0
looking_up	Gaze directed upward	$0^\circ \pm 2^\circ$	$+10^\circ \pm 3^\circ$	0.7-1.0
looking_down	Gaze directed downward	$0^\circ \pm 2^\circ$	$-10^\circ \pm 3^\circ$	0.7-1.0
engaged	Generally attentive	$0^\circ \pm 5^\circ$	$0^\circ \pm 5^\circ$	0.7-1.0
partial_engaged	Partially attentive	Variable	Variable	0.7-1.0
sleepy	Eyes closing, drowsy	$0^\circ \pm 5^\circ$	$-5^\circ \pm 3^\circ$	0.0-0.3
distracted_by_multi_face	Multiple faces detected	Variable	Variable	0.5-1.0
no_face	No face detected	N/A	N/A	N/A
unknown	Uncertain state	Variable	Variable	0.5-1.0

Data Format

Metadata Structure (JSONL)

Each line in `metadata.jsonl` contains one JSON object with the following fields:

```
{
  "frame_path": "frames/synthetic_00000.jpg",
  "gaze_yaw": -15.76,
  "gaze_pitch": -4.40,
  "head_yaw": -6.96,
  "head_pitch": -1.15,
  "eyes_open_prob": 0.788,
  "label": "looking_left",
  "confidence": 0.789,
  "meta": {
    "source": "synthetic",
    "index": 0
  }
}
```

Field Descriptions

Field	Type	Description	Range
frame_path	string	Relative path to image file	-
gaze_yaw	float	Horizontal gaze angle (degrees)	-30° to +30°
gaze_pitch	float	Vertical gaze angle (degrees)	-20° to +20°
head_yaw	float	Horizontal head rotation (degrees)	-30° to +30°
head_pitch	float	Vertical head rotation (degrees)	-20° to +20°
eyes_open_prob	float	Probability eyes are open	0.0 to 1.0
label	string	Attention state classification	See labels above
confidence	float	Label confidence score	0.7 to 0.95
meta	object	Additional metadata	-

Sample Data Examples

Example 1: Focused Student

```
{
  "frame_path": "frames/synthetic_00001.jpg",
  "gaze_yaw": 2.51,
  "gaze_pitch": 4.10,
  "head_yaw": -5.49,
  "head_pitch": -5.07,
  "eyes_open_prob": 0.86,
  "label": "focused",
  "confidence": 0.71
}
```

Example 2: Looking Left

```
{
  "frame_path": "frames/synthetic_00000.jpg",
  "gaze_yaw": -15.76,
  "gaze_pitch": -4.40,
  "head_yaw": -6.96,
  "head_pitch": -1.15,
  "eyes_open_prob": 0.79,
  "label": "looking_left",
  "confidence": 0.79
}
```

Example 3: Sleepy Student

```
{
  "frame_path": "frames/synthetic_00042.jpg",
  "gaze_yaw": 1.23,
  "gaze_pitch": -6.45,
  "head_yaw": 2.11,
  "head_pitch": -7.89,
  "eyes_open_prob": 0.15,
  "label": "sleepy",
  "confidence": 0.85
}
```

```
"confidence": 0.88
}
```

Files That Use This Dataset

1. Training Script

File: ml/train.py

Usage: Loads the dataset and trains the attention detection model

```
# Load samples from metadata.jsonl
samples = load_samples(cfg.data_path)
train_samples, val_samples = train_val_split(samples)

# Preprocess for training
train_data = preprocess_samples(train_samples, cfg.image_size)
```

Key Functions:

- `load_samples()` - Reads metadata.jsonl
- `preprocess_samples()` - Converts to model-ready format
- `train_val_split()` - Splits into training/validation sets

2. Dataset Module

File: ml/dataset.py

Usage: Provides data loading and preprocessing utilities

3. Evaluation Script

File: ml/eval.py

Usage: Evaluates model performance on the dataset

4. Data Ingestion Script

File: ml/data_ingest.py

Usage: Generates and normalizes the dataset

Key Function: `generate_synthetic_samples(count, output_dir)`

Training Process

Step 1: Generate Dataset

```
python ml/data_ingest.py ingest synthetic_demo --limit 1000
```

Step 2: Train Model

```
python ml/train.py \  
  --data data/processed/synthetic_demo/metadata.jsonl \  
  --epochs 10 \  
  --batch-size 32
```

Step 3: Export Model

```
python ml/export_model.py \  
  --model artifacts/focusmate_model/saved_model \  
  --out frontend/public/models/focusmate-attention \  
  --format tfjs
```

Model Architecture

Base Model

MobileNetV3-Small (lightweight, optimized for mobile/web)

Input Specifications

- **Size:** 160×120 pixels
- **Format:** RGB (3 channels)
- **Normalization:** Pixel values divided by 255.0

Output Heads (Multi-task Learning)

1. **Gaze Yaw** - Horizontal gaze angle
2. **Gaze Pitch** - Vertical gaze angle
3. **Head Yaw** - Horizontal head rotation
4. **Head Pitch** - Vertical head rotation
5. **Eyes Open Probability** - Eye openness score
6. **Label Probabilities** - 11-class attention classification

Model Size

- **Uncompressed:** ~10MB
 - **Quantized:** <5MB
 - **Format:** TensorFlow.js (for browser deployment)
-

Alternative Datasets (Production Use)

For production deployment, the following public datasets are recommended:

1. GazeCapture (MIT)

- **URL:** <https://gazecapture.csail.mit.edu/>
- **License:** Research-only (request access)

- **Size:** 2.5M frames from 1,474 subjects
- **Notes:** Mobile-focused, on-screen vs off-screen attention

2. MPIIGaze

- **URL:** <https://www.mpi-inf.mpg.de/departments/computer-vision-and-machine-learning/research/gaze-estimation/mpiigaze>
- **License:** Non-commercial (request access)
- **Size:** 213,659 images from 15 subjects
- **Notes:** Laptop gaze with head-pose diversity

3. Columbia Gaze

- **URL:** http://www.cs.columbia.edu/CAVE/databases/columbia_gaze/
- **License:** Non-commercial (direct download)
- **Size:** 5,880 images from 56 subjects
- **Notes:** Discrete left/right/up/down gaze samples

4. ETH-XGaze

- **URL:** <https://ait.ethz.ch/projects/2020/xgaze/>
- **License:** Research-only (request access)
- **Size:** 1M+ images from 110 subjects
- **Notes:** Large multi-view head-pose dataset

5. OpenEDS (Facebook/Meta)

- **URL:** <https://research.fb.com/openeds-2020-challenge/>
- **License:** CC BY-NC 4.0 (registration required)
- **Size:** 12,759 sequences
- **Notes:** Eye segmentation and eye-openness cues

Dataset Statistics

Label Distribution (Synthetic Dataset)

The synthetic dataset generates approximately equal distribution across all 11 labels:

- Each label: ~9% of total samples
- Ensures balanced training across all attention states

Angle Distributions

Gaze Angles:

- Yaw range: -30° to $+30^\circ$
- Pitch range: -20° to $+20^\circ$
- Gaussian distribution around label-specific means

Head Pose:

- Yaw range: -30° to +30°
- Pitch range: -25° to +25°
- Correlated with gaze direction but with added variance

Eye Openness:

- Focused/Engaged: 0.7-1.0
 - Sleepy: 0.0-0.3
 - Other states: 0.5-1.0
-

Configuration

Training Configuration

File: ml/config.py

```
@dataclass
class TrainingConfig:
    data_path: Path
    output_dir: Path = Path("artifacts/model")
    epochs: int = 5
    batch_size: int = 32
    learning_rate: float = 2e-4
    image_size: tuple[int, int] = (120, 160)
    label_smoothing: float = 0.05
    num_workers: int = 4
```

Quality Assurance

Data Validation

- All gaze/head angles within realistic ranges
- Eye openness probabilities between 0.0 and 1.0
- Confidence scores between 0.7 and 0.95
- All image files exist and are readable

Label Consistency

- Gaze angles match expected ranges for each label
 - Eye openness appropriate for attention state
 - Head pose correlated with gaze direction
-

Future Enhancements

Planned Improvements

1. **Real Data Collection:** Implement local calibration capture tool
2. **Public Dataset Integration:** Add adapters for GazeCapture, MPIIGaze, etc.

3. **Data Augmentation:** Add rotation, brightness, blur variations
4. **Active Learning:** Collect and label edge cases from production
5. **Multi-face Support:** Enhanced distraction detection

Adapter Implementation

To add support for public datasets, implement in `ml/data_ingest.py`:

1. Parse dataset-specific format
 2. Extract gaze angles, head pose, eye openness
 3. Map to Focus Mate attention labels
 4. Write to unified JSONL format
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References

Code Files

- `ml/data_ingest.py` - Dataset generation and ingestion
- `ml/train.py` - Model training pipeline
- `ml/dataset.py` - Data loading utilities
- `ml/config.py` - Training configuration
- `ml/labels.py` - Label definitions and mappings

Documentation

- `ml/README.md` - ML pipeline overview
 - `docs/FEATURES.md` - Complete feature documentation
 - `TRAINING_GUIDE.md` - Step-by-step training guide
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Contact & Support

For questions about the dataset or training process:

1. Review the documentation in `ml/README.md`
 2. Check the training guide in `TRAINING_GUIDE.md`
 3. Examine the code in `ml/data_ingest.py`
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Project: Focus Mate Virtual Classroom